

Object recognition

It is very difficult for a person to look at a washing machine and confuse it with a television. Though humans can recognize a large number of objects easily, the recognition techniques are in their infancy for computers. Object recognition has its roots in artificial

intelligence, pattern recognition and is also one of the driving factors for the image registration process. Interestingly enough the technology has been imagined repeatedly by filmmakers, right from *The Terminator* to *Transformers*.

Most object recognition methods heavily depend on objects that are already known. The larger the number of objects known to the system, better the chances of a correct object recognition. Let us first have a closer look at what recognition methods can act as hints in forming the complete understanding of the object. We would start with simple detection techniques and advance as we move forward.



Remember how Terminator recognised objects such as guns and even the jacket sizes

Looking for gradients

Any curved surface reflects light as gradients as reflection varies in a continuous fashion across the surface. The gradient produced by a cylindrical object is different from that of a conical object. Observing images received via the camera to study the gradients helps in figuring out the overall structure of the object. The variation of gradients based on the lighting conditions for each type of shape which build the object altogether pose a major challenge here.

Comparing grayscale versions

Grayscale versions of images are easier to compare than the colored versions. Think of it like this: pictures of an pencil box when taken under green light looks different than picture of the same pencil box taken in blue light. But when you convert both the images into grayscale (i.e. black and white) the pictures do not look that much different. While to the human eye, the difference is trivial even in color, it makes a significant difference to a computer-based algorithm.